

READING PASSAGE 3

3D Printing and the Future of Manufacturing

A In recent years, 3d printing and the future of manufacturing has attracted considerable attention from both the academic community and the general public. This growing interest reflects a broader recognition of the importance of this topic for understanding some of the most pressing challenges facing contemporary society.

B Historically, scholarly understanding of this topic was limited by both theoretical frameworks and available methodologies. Early researchers operated within relatively narrow conceptual boundaries, and their findings, while groundbreaking for their time, often failed to capture the full complexity of the phenomena they sought to explain. The development of new research tools and analytical techniques in the latter half of the twentieth century fundamentally transformed the landscape, enabling scholars to investigate questions that had previously been considered beyond the reach of empirical inquiry.

C The mechanisms underlying these phenomena operate at multiple levels of analysis, from the micro-level of individual decision-making to the macro-level of institutional and structural change. At the individual level, cognitive processes, emotional responses, and social influences shape the choices that people make. At the organisational level, formal rules, informal norms, and power dynamics govern collective action. At the societal level, historical legacies, cultural values, and political structures create the broad context within which lower-level processes unfold.

D A substantial body of empirical evidence supports these theoretical frameworks. A landmark study conducted by researchers at leading international institutions analysed data spanning a thirty-year period and involving participants from more than forty countries. The findings revealed a consistent pattern of results that held across diverse cultural and economic contexts, suggesting that the identified mechanisms operate with considerable universality. The magnitude of the observed effects was particularly striking, with the key variables accounting for a substantial proportion of the total variance in outcomes.

E Several unresolved debates continue to animate scholarly discussion in this field. Perhaps the most prominent concerns the relative importance of structural versus agentic factors in determining outcomes. While some researchers emphasise the constraining power of institutional and economic structures, others highlight the capacity of individuals and groups to exercise agency and effect change even within seemingly restrictive contexts. This debate has significant practical implications, as it shapes recommendations for policy interventions and institutional reforms.

F The future trajectory of research in this area will likely be shaped by several converging trends. Advances in computational methods and data science are opening up new analytical possibilities, enabling researchers to detect patterns and test hypotheses at scales that were previously unimaginable. Simultaneously, growing recognition of the importance of interdisciplinary collaboration is breaking down traditional barriers between fields and fostering innovative syntheses of diverse theoretical perspectives.

G In conclusion, the study of 3d printing and the future of manufacturing represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Questions 1–2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 1 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.
- 2 The author notes that researchers have called for more _____ approaches to investigation.

Questions 3–4

Choose the correct letter, A, B, C or D.

- 3 The author suggests that early studies in this field were limited by:
 - A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone
- 4 The passage indicates that critics have raised concerns about:
 - A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Question 5

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 5 All critics agree that the research findings are universally applicable.

Question 6

Choose the correct letter, A, B, C or D.

- 6 What does the passage identify as a significant challenge to current understanding?
 - A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations

Question 7

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

7 Policy-makers have fully implemented all research recommendations.

Question 8

Choose the correct letter, A, B, C or D.

- 8 According to the text, what role do structural factors play?
- A. minimal impact
 - B. creating underlying conditions for action
 - C. determining individual choices directly
 - D. irrelevant to outcomes

Question 9

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 9 Contemporary understanding of this topic is more sophisticated than it was fifty years ago.

Question 10

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 10 The author notes that researchers have called for more _____ approaches to investigation.

Question 11

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 11 Early researchers had access to the same technologies as modern scholars.

Question 12

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 12 The landmark study analysed data spanning a period of _____ years.

Question 13

Choose the correct letter, A, B, C or D.

- 13 The meta-analyses cited in the passage reveal that:
- A. excessive research funding
 - B. translating academic findings into practical policy

C. lack of public interest

D. too many international collaborations